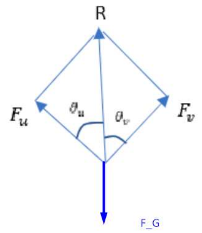
--- HW1 Content ---

Resultant Forces



$$\uparrow^{+} + \Sigma F_y = 0, \quad \rightarrow^{+} \Sigma F_x = 0$$

Linear Solve for:

$$(1)\ \ -F_G + F_u \sin\theta + F_v \sin\theta = 0$$
$$(2)\ \ F_u \cos\theta = F_v \cos\theta$$

Robot Driving up a Slopez

Downhill Drive: $F_{net} = \frac{\tau}{R} + mg\sin(\theta) - mg\cos(\theta)\, C_r(v)$

Upward Climb: $F_{net} = \frac{\tau}{R} - mg\sin(\theta) - mg\cos(\theta)\, C_r(v)$

$$F = ma, \qquad t = \frac{v}{a}$$

--- HW2 Content ---

Velocity Equations in a Rotating Frame

$$\dot{x}' = \dot{x_o}' - \sin(\theta) * \dot{\theta} * x + \cos(\theta) * \dot{x} - \cos\theta * \dot{\theta} * y - \sin\theta * \dot{y}$$

$$\dot{y}' = \dot{y_o}' + \cos(\theta) * \dot{\theta} * x + \sin(\theta) * \dot{x} - \sin(\theta) * \dot{\theta} * y + \cos(\theta) * \dot{y}$$

Global-Local Displacement, Final Position

$$x_A' = x_o' + x_n' + x_A * \cos(\theta) - y_A * \sin(\theta)$$

$$y_A' = y_o' + x_n' + x_A * \sin(\theta) + y_A * \cos(\theta)$$

Where $\Sigma\theta = \circlearrowright + \theta$

Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

--- HW3 Content ---

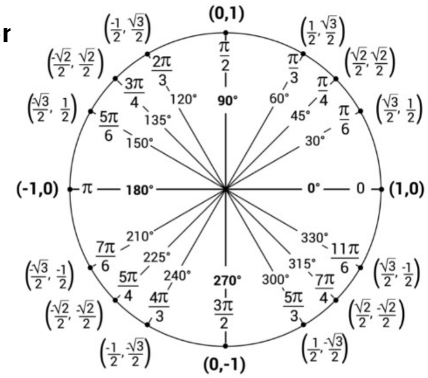
Basic Motion Definitions

$$\dot{x} = \frac{\text{distance}}{\text{time}}$$

$$\dot{\theta} = \frac{\text{rad}}{\text{time}}$$

Trajectory - Arc Motior

$\dot{\phi}_R$	$\dot{\phi}_L$	FWD
$\dot{\phi}_R$	-$\dot{\phi}_L$	RIGHT
-$\dot{\phi}_R$	-$\dot{\phi}_L$	REV
-$\dot{\phi}_R$	$\dot{\phi}_L$	LEFT
$\dot{\phi}_R$	-$\dot{\phi}_L$	CCW
-$\dot{\phi}_R$	$\dot{\phi}_L$	CW



Solve for $\dot{\phi}_R$:

$$\frac{\text{distance}}{\text{time}} = \text{Wheel radius R} * \dot{\phi}_R$$

$$\dot{x} = \frac{\frac{2\pi r}{4}}{time}\ \text{m/s}$$

$$\dot{\theta} = \frac{\theta_{\text{final}} - \theta_{\text{initial}}}{time}\ \text{rad/s (Use Unit Circle)}$$

To Solve for Phi Dot L & R, Linear Solve the Following:

$$\dot{x} = \frac{R}{2}(\dot{\phi}_L + \dot{\phi}_R)\ \text{m/s}$$

$$\dot{\theta} = \frac{R}{2L}(\dot{\phi}_R - \dot{\phi}_L)\ \text{rad/s}$$

--- HW4 Content ---

Trajectory - Mathematical Function

Degrees→ radians
$\times$ by $\frac{\pi}{180}$
Radians→ degrees
$\times$ by $\frac{180}{\pi}$

Given $y' = \sqrt[3]{x}$ from x'=1 to x'=1.5 m:

$$\theta_{\text{final (2)}} = \arctan\left(\frac{d}{dx}(y')|_{x=1.5}\right) \text{ degrees, convert to rad after}$$

$$\theta_{\text{initial (1)}} = \arctan\left(\frac{d}{dx}(y')|_{x=1}\right) \text{ degrees, convert to rad after}$$

$$\dot{\theta} = \frac{\theta_{\text{final}} - \theta_{\text{initial}}}{time}\ rad/s$$

$$\text{distance} = \frac{\sqrt{(y_2' - y_1')^2 + (x_2' - x_1')^2}}{2\sin\left(\frac{|\theta_2 - \theta_1|}{2}\right)}$$

$$\dot{x} = \frac{\text{Distance} \times |\theta_2 - \theta_1|}{t}$$

Same steps for finding Phi Dot as in HW3 Section